Calvin Trees – Interactive Campus Arboretum

Vision Statement

Our goal is to continue development and complete production of Professor Norman's existing Calvin Trees app. The vision is to engage various demographics (ranging from college students to campus visitors of all ages) with our campus, specifically the natural side of Calvin University. We will accomplish this by polishing existing UI elements, finishing implementation of partially completed features, and introduce brand new features altogether. Key ethical and design norms for our project include cultural appropriateness and aesthetic norms. It is imperative that we develop something that can foster interest in our campus tree ecosystem without disturbing the ecosystem or cause distraction and, consequently, detract from the entire purpose of engaging with nature. Moreover, the app and new functionalities need to be intuitive and easy to understand to make the core purpose of exploring Calvin University's trees enjoyable.

Normative and Ethical Consideration

Social

A potential challenge we could have with this norm is that our project could take away from engagement with Calvin University's natural environment. Creating a mobile application that distracts from truly appreciating our campus' surroundings. We can combat this by implementing simple UI and focusing on guiding without too many pop ups and visual interruptions.

Aesthetics

We need to ensure that our application is aesthetically pleasing to engage with, and intuitive enough to allow seamless usage and maximize engagement with our campus environment. Our team can accomplish this by adhering to key UI design principles such as maintaining consistency, utilizing contrast efficiently, making the application's features accessible and simple, and conduct user tests to ensure we are not biased.

Cultural Appropriateness

Our app should encourage meaningful interaction with the environment while ensuring that Calvin University's ecosystem remains undisturbed. We can achieve this by incorporating guidance and gentle reminders for users. These could include informative messages that promote responsible behavior, such as properly disposing of trash and respecting natural spaces.

Background

Traditional arboretum and environmental engagement platforms are often designed for large botanical gardens, national parks, or funded institutions with significant budgets and staff. Examples include iNaturalist which provides powerful tools for mapping, species tracking, and ecological data collection. While robust in functionality, these are typically built for broader scientific or large-scale public engagement and requires substantial ongoing maintenance and funding.

For smaller campus-based engagement like with Calvin University, these solutions may be too complex, expensive, and/or misaligned with this institution's goals. The primary objective is to foster student engagement, environmental awareness, and appreciation of the campus' natural ecosystem. A generalized platform may not accomplish this. Our application is highly focused and accessible, especially for mobile users, because of our progressive web application implementation. Rather than being a scientific database, Calvin Trees focuses on proximity-based notifications, randomly generated waking tours, and simple interactive guidance. Our implementation ensures minimal ongoing maintenance costs and requirements.

Development Process

We are storing source code on GitHub, meeting with our team and professor weekly (Thursday), and commit 6 hours weekly as a minimum.

- GitHub
  - Production Branch → currently gets deployed via Firebase Hosting
  - Dev Branch → individuals branch from here and use PRs for merging code
  - Quality Checks → implement linter and code reviews for coding conventions
- Planning and Organization
  - Notion → organize schedules and broader backlog tasks
  - GitHub Projects → organize tasks with greater specificity, delegate tasks

Contributions

- Sam
  - Fixed numerous bugs including a duplicate search results bug, fixed the marker and location implementation, and assisted in enhanced error handling

- Peter
    - Investigated initial implementation, fixed layout and modals for individual trees, updated our tree data
- Alim
    - Built function CRUD for tree data and admin capability, full testing coverage, and implementation of two tour mode

Success Criteria

- Development of fully functional PWA that reliably delivers all core features including:
    - Interactive mapping
    - Proximity-based notifications
    - Information tree components
    - Guided walking tours

Approach and Implementation

Source Control

Accomplished through Git, primarily working off of the dev branch and merging changes through PRs (which were generally scoped to one behavior or documentation area at a time.

Editor and Tooling

TypeScript strict mode, ESLint with Angular and TypeScript plugin sets through npm run lint.

Layered Architecture

UI components were focused on view state and presentation. Persistence, parsing, and searching lives in the TreeService. Admin page never reaches into localStorage directly. A full mermaid diagram depicting our architecture can be found on our team website in the Reports page. Additionally, we implemented ShowTreeMarkersComponent which is the single component responsible for rendering tree marker layers. The home page configures multiple instances including trees, search results, and tour targets instead of duplicating MapLibre layer code.

Testing

Include 201 built-in tests across 7 spec files, every change was proceeded by running full testing suite to ensure functionality. Included peer reviewing between group members and physical testing by walking around campus.

Challenges

- Layout/Modal Issues
  - Implementing tree markers, map layer, and tree pop-up modals proved challenging. Had issues with components populating UI and resolving any overflow. Required slowing down, working with MapLibre documentation, and essentially recreating that component.
- Cross-Device Compatibility
  - All developers had iOS hardware during development. Ensuring Android functionality required usage of an emulator, which limited our ability to do physical testing. Many features required tweaks after development and testing for iOS environment to ensure they worked correctly for Android.
- PWA Offline Capability
  - Current base level caching and offline capability sufficient. Future work requires finding solution to cache, at the very least, the map area around Calvin for PWA. Currently, hard reloading the application does not consistently populate the map boundary containing tree markers.

Results

Demo is included in the team website in Reports page.

Successfully implemented interactive mapping, proximity-based notifications, information-based tree components, and two guided walking tours. All group members contributed equally in terms of developing features, polishing existing codebase, testing, producing documentation for milestones, and for demoing our final product. Communication between team members was efficient. Possible feature work falls into two categories.

Features/Improvements:

- Gamification (achievements and milestones to promote recurring usage)
- Optimize for larger tree data sets
- Optimize geolocation frequency
- Introduce better Android build support
- Enhance offline map caching
- Push notifications

Engagement:

- Use flyers and website to promote the platform
- Integrate into campus orientation
- Promote social sharing

Ultimately, an amazing milestone would be to deploy in the App Store and on Google Play.

Conclusion

Our team was able to deliver an MVP that is presentable and usable by the general Calvin community. The web app is deployed here and can be saved as a shortcut to home screens for use as a PWA.

Each team member contributed meaningfully to the above work. There were no designated roles (e.g. UI design, backend design, etc.). Instead, we all contributed to the accomplishments above equally and all spent time testing our application rigorously.

The development team firmly believes that they have expanded upon the existing project and contributed meaningfully given the pivot following the fall semester. The team learned the importance of deterministic and scoped requirements gathering to ensure productive usage of development time following pivoting projects. Multiple features and polish have been successfully implemented, and a strong foundation has been created for any future development. Calvin Trees provides a platform to enhance engagement with Calvin University's diverse tree ecosystem without detracting from the primary goal of such interaction, that being appreciating the environment and fostering respect for God's creation. The team delivered something that balances that with an application that is aesthetically pleasing and intuitive to use.